

Double Machine Learning-Based Programme Evaluation Under Unconfoundedness

Michael C. Knaus · The Econometrics Journal (2022)

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Do Swiss active labour market programs cause unemployed workers to find more employment?

And does the effect differ by who you are?

Replicating: Table 4 — Group Average Treatment Effects

4 Programs studied:

- Job Search
- Vocational Training
- Computer Training
- Language Courses

vs. No Program (control)

Heterogeneity across:

- Gender
- Foreigner Status
- Employability Level

Swiss Active Labour Market Policy Dataset

100,120

Raw Dataset Obs.

62,465

Observations
(paper: 62,497)

Program	N
No program	47,620
Job search	11,610
Vocational	858
Computer	905
Language	1,504

45

Covariates

31

Months of
outcome window

Cleaning:

- Keeps German-speaking cantons
- Drops employment and personality programs
- Y = total employed months
- W = program type
- X = 45 covariates/controls

What is Double Machine Learning?

The core idea: use machine learning to strip out confounding, then do causal inference on what's left.

1

Predict outcomes from X

A random forest predicts each person's months employed using their demographics, work history, and region — ignoring treatment entirely. This is the outcome nuisance model $\mu(w, X)$.

2

Predict treatment from X

A second random forest predicts the probability of ending up in each program, given those same characteristics. This is the propensity score $\hat{e}_w(X)$. Removes selection bias.

3

Run OLS on residuals/ Compute doubly robust pseudo-outcomes, then regress those pseudo-outcomes on subgroup indicators.

Subtract both predictions. What remains is variation in outcomes and treatment that X cannot explain. OLS on these residuals gives a clean causal estimate. No linearity assumed in step 1 or 2.

Cross-fitting (5 folds): nuisance models never predict their own training data — prevents overfitting from contaminating the causal step.

DML in Class vs. Knaus (2022)

Same framework but different complexity. In class we used the partially linear model (PLR); Knaus extends to multiple treatments with a doubly robust score.

In Class — Partially Linear Model

$$\begin{aligned} Y &= D\theta + g(X) + \varepsilon \\ D &= m(X) + v \end{aligned}$$

Nuisance 1: $\hat{g}(X) = E[Y | X] \rightarrow$ LassoCV or RF

Nuisance 2: $\hat{m}(X) = E[D | X] \rightarrow$ LassoCV or RF

Residuals: $\tilde{Y} = Y - \hat{g}(X), \tilde{D} = D - \hat{m}(X)$

Final step: $\hat{\theta} = \Sigma \tilde{D} \tilde{Y} / \Sigma \tilde{D}^2$ (OLS residual-on-residual)

Treatment: Binary (0/1)

Covariates: 7 (age, educ, earnings + interactions)

Knaus (2022) — AIPW / Multiple Treatments

$$\hat{\Gamma}_{i,w} = \hat{\mu}(w, X_i) + \frac{D_i(w) (Y_i - \hat{\mu}(w, X_i))}{\hat{e}_w(X_i)}$$

Nuisance 1: $\hat{\mu}(w, X) = E[Y | W=w, X] \rightarrow$ Honest RF

Nuisance 2: $\hat{e}_w(X) = P[W=w | X] \rightarrow$ Honest RF

Score: Doubly robust APO score $\Gamma_{i,w}$

Final step: OLS of $\hat{\Delta}$ on subgroup dummies (Table 4)

$$\hat{\Delta}_{i,w,w'} = \hat{\Gamma}_{i,w} - \hat{\Gamma}_{i,w'}$$

Treatment: 5-way (no prog / job / voc / comp / lang)

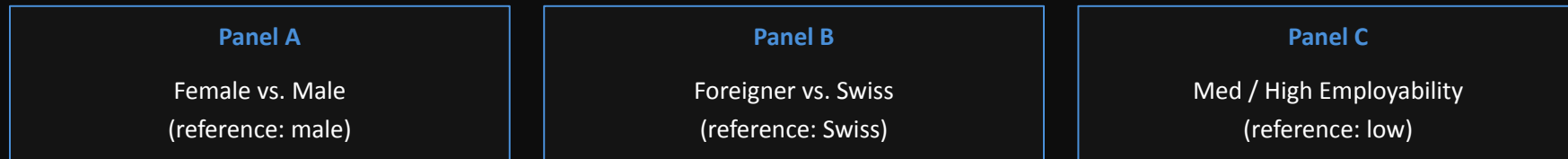
Covariates: 45 (demographics, work history, etc.)

Table 4 — Group Average Treatment Effects

The DML pseudo-outcome $\Delta_i = \Gamma_{i,w} - \Gamma_{i,o}$ is regressed via OLS on subgroup dummies. Each coefficient = how much that group's treatment effect differs from the reference group.



Three subgroup panels:



Why Can't We Just Compare Averages?

The Problem

- Caseworkers — not random chance — decide who enters which program
- People with better job prospects are more likely to get hard-skill training
- 45 confounders: demographics, work history, caseworker traits, regional conditions
- Naive comparison = selection bias, not a causal effect

DML Fix

- Random forest controls for all 45 covariates without assuming linearity
- Propensity score soaks up selection into treatment — no manual modeling
- Doubly robust: result holds even if one nuisance model is slightly wrong
- The final OLS step allows standard inference after the DML adjustment, under unconfoundedness.

My Results vs. Knaus (2022) — Table 4

Mine

McKnaus

Group average treatment effects

	Job search (1)	Vocational (2)	Computer (3)	Language (4)
<i>Panel A: Female</i>				
Constant	-1.24*** (0.17)	3.67*** (0.56)	2.34*** (0.59)	3.44*** (0.46)
Female	0.65** (0.25)	-1.01 (0.86)	2.65*** (0.84)	-1.89** (0.77)
<i>Panel B: Foreigner</i>				
Constant	-1.18*** (0.16)	2.46*** (0.53)	3.79*** (0.50)	3.74*** (0.51)
Foreigner	0.62** (0.26)	2.09** (0.89)	-0.79 (0.91)	-3.12*** (0.71)
<i>Panel C: Employability</i>				
Constant	-0.10 (0.33)	5.44*** (1.04)	5.81*** (1.08)	2.97*** (0.84)
Medium employability	-0.95*** (0.36)	-2.41** (1.16)	-2.59** (1.19)	-0.48 (0.95)
High employability	-1.57*** (0.50)	-4.40*** (1.49)	-3.92** (1.67)	-0.02 (1.45)
F-statistic	5.37***	4.40**	3.23**	0.17

Notes: OLS coefficients and heteroscedasticity robust standard errors in parentheses. Regressions use the DML pseudo-outcome. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 4. Group average treatment effects.

	Job search (1)	Vocational (2)	Computer (3)	Language (4)
<i>Panel A: Female</i>				
Constant	-1.29*** (0.17)	3.82*** (0.55)	2.33*** (0.60)	3.40*** (0.46)
Female	0.60** (0.25)	-1.27 (0.87)	2.49*** (0.85)	-1.97** (0.77)
<i>Panel B: Foreigner</i>				
Constant	-1.27*** (0.16)	2.48*** (0.53)	3.75*** (0.50)	3.56*** (0.52)
Foreigner	0.70*** (0.26)	2.17** (0.90)	-0.88 (0.93)	-2.84*** (0.72)
<i>Panel C: Employability</i>				
Constant	-0.18 (0.33)	5.48*** (1.04)	5.76*** (1.10)	2.61*** (0.85)
Medium employability	-0.93*** (0.36)	-2.41** (1.16)	-2.67** (1.20)	-0.18 (0.96)
High employability	-1.50*** (0.50)	-4.42*** (1.52)	-3.59** (1.69)	0.59 (1.48)
F-statistic	5.04***	4.31**	2.96*	0.18

Notes: This table shows OLS coefficients and their heteroscedasticity robust standard errors (in parentheses) of regressions run with the pseudo-outcome defined as described in Section 3.3. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

(1) Signs and significance patterns are very close. (2) Magnitudes are close but don't match because I ran a fixed 500-tree Random Forest and our dataset observations were slightly off. (3) We see from the table that Job Search is weak/negative while the others show larger positive employment gains

INTERPRETATION

- 1 Job search program: about 1.2 fewer employed months for the male reference group.
- 2 Vocational/computer/language programs: roughly 3 to 4 more employed months for many reference groups.
- 3 Heterogeneity matters: one average effect hides who gains and who loses.

CHALLENGES

- 1 Reconstructing the sample from raw data.
- 2 Assigning pseudo-start dates to no-program observations.
- 3 Balancing computation time against fully tuned forest specifications.

Three Things I Learned

01

DML = ML + causal inference, not just prediction

The machine learning step handles the messy confounding. The doubly robust score handles identification. The final OLS step is clean enough for standard inference. Neither step alone would work.

02

One average effect hides the real story

Job search is weak or negative overall, but the size of that effect still differs by subgroup. Hard-skill programs generally help, especially for lower-employability workers, so policy should target programs rather than treat everyone the same.

03

Reproducibility is harder than estimation

The code is the easy part. Sample restrictions, pseudo-start date assignment, and ML tuning choices all move the numbers. Estimation code without data and documentation is not reproducible.